

VULNERABILITY ANALYSIS - ENUMERATION

Target Machine: **Metasploitable**

Target IP: **192.168.56.110**

```
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:0d:2a:ac
          inet addr:192.168.56.110  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe0d:2aac/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:2 errors:0 dropped:0 overruns:0 frame:0
          TX packets:30 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:1188 (1.1 KB)  TX bytes:3924 (3.8 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:109 errors:0 dropped:0 overruns:0 frame:0
          TX packets:109 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:27661 (27.0 KB)  TX bytes:27661 (27.0 KB)

msfadmin@metasploitable:~$ _
```

Section A – Basic Enumeration

Challenge 1 - NetBIOS Enumeration

Command: nbtstat -a 192.168.56.110

Command in **Kali**: nbtscan 192.168.56.110

```
(kali㉿kali)-[~]
$ nbtscan 192.168.56.110

Doing NBT name scan for addresses from 192.168.56.110

IP address      NetBIOS Name    Server    User          MAC address
-----
-
192.168.56.110  METASPLOITABLE <server>  METASPLOITABLE 00:00:00:00:00:00
0

(kali㉿kali)-[~]
$
```

Challenge 2 – Fast Nmap Scan

Command: `nmap -F 192.168.56.110`

```
(kali㉿kali)-[~]
└─$ nmap -F 192.168.56.110
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-01 11:24 +08
Nmap scan report for 192.168.56.110
Host is up (0.0039s latency).
Not shown: 82 filtered tcp ports (no-response)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
513/tcp   open  login
514/tcp   open  shell
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
8009/tcp  open  ajp13

Nmap done: 1 IP address (1 host up) scanned in 1.88 seconds
```

Challenge 5 - TTL OS Fingerprinting

TTL values reference:

TTL OS Guess

64 Linux / Unix

128 Windows

255 Cisco / BSD

```
(kali㉿kali)-[~]
└─$ ping -c 4 192.168.56.110
PING 192.168.56.110 (192.168.56.110) 56(84) bytes of data.
 64 bytes from 192.168.56.110: icmp_seq=1 ttl=255 time=1.80 ms
 64 bytes from 192.168.56.110: icmp_seq=2 ttl=255 time=1.89 ms
 64 bytes from 192.168.56.110: icmp_seq=3 ttl=255 time=1.92 ms
 64 bytes from 192.168.56.110: icmp_seq=4 ttl=255 time=1.81 ms

— 192.168.56.110 ping statistics —
 4 packets transmitted, 4 received, 0% packet loss, time 3010ms
 rtt min/avg/max/mdev = 1.800/1.853/1.917/0.050 ms
```

Challenge 7 – SMTP VRFY/EXPN

Command: nc 192.168.56.110 25

```
(kali㉿kali)-[~]  
$ nc 192.168.56.110 25  
220 metasploitable.localdomain ESMTF Postfix (Ubuntu)  
VRFY root  
252 2.0.0 root  
QUIT  
221 2.0.0 Bye
```

Challenge 9 – FTP Banner

Command: nc 192.168.56.110 21

```
(kali㉿kali)-[~]  
$ nc 192.168.56.110 21  
220 (vsFTPd 2.3.4)  
█
```

Challenge 10 – Anonymous FTP Login

Command 192.168.56.110 21

```
(kali㉿kali)-[~]  
$ ftp 192.168.56.110 21  
Connected to 192.168.56.110.  
220 (vsFTPd 2.3.4)  
Name (192.168.56.110:kali): anonymous  
331 Please specify the password.  
Password:  
230 Login successful.  
Remote system type is UNIX.  
Using binary mode to transfer files.  
ftp> ls  
229 Entering Extended Passive Mode (|||58200|).  
150 Here comes the directory listing.  
226 Directory send OK.  
ftp> exit  
221 Goodbye.
```

SECTION B — INTERMEDIATE ENUMERATION

Challenge 11 – SMB NSE Enumeration

Command: `nmap --script smb-os-discovery -p445 192.168.56.110`

```
(kali㉿kali)-[~]
$ nmap --script smb-os-discovery -p445 192.168.56.110
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-01 12:08 +08
Nmap scan report for 192.168.56.110
Host is up (0.00094s latency).

PORT      STATE SERVICE
445/tcp   open  microsoft-ds

Host script results:
| smb-os-discovery:
|   OS: Unix (Samba 3.0.20-Debian)
|   Computer name: metasploitable
|   NetBIOS computer name:
|   Domain name: localdomain
|   FQDN: metasploitable.localdomain
|_  System time: 2026-05-01T00:08:13-04:00

Nmap done: 1 IP address (1 host up) scanned in 0.30 seconds
```

Command: `nmap --script smb-enum-users -p445 192.168.56.110`

```
(kali㉿kali)-[~]
$ nmap --script smb-enum-users -p445 192.168.56.110
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-01 12:08 +08
Nmap scan report for 192.168.56.110
Host is up (0.00063s latency).

PORT      STATE SERVICE
445/tcp   open  microsoft-ds

Host script results:
| smb-enum-users:
|   METASPLOITABLE\backup (RID: 1068)
|     Full name: backup
|     Flags: Normal user account, Account disabled
|   METASPLOITABLE\bin (RID: 1004)
|     Full name: bin
|     Flags: Normal user account, Account disabled
|   METASPLOITABLE\bind (RID: 1210)
|     Flags: Normal user account, Account disabled
|   METASPLOITABLE\daemon (RID: 1002)
|     Full name: daemon
|     Flags: Normal user account, Account disabled
|   METASPLOITABLE\dhcp (RID: 1202)
|     Flags: Normal user account, Account disabled
|   METASPLOITABLE\distccd (RID: 1222)
|     Flags: Normal user account, Account disabled
|   METASPLOITABLE\ftp (RID: 1214)
|     Flags: Normal user account, Account disabled
|   METASPLOITABLE\uucp (RID: 1020)
|     Full name: uucp
|     Flags: Normal user account, Account disabled
|   METASPLOITABLE\www-data (RID: 1066)
|     Full name: www-data
|     Flags: Normal user account, Account disabled
|_

Nmap done: 1 IP address (1 host up) scanned in 0.38 seconds
```

Challenge 12 – Enum4linux

Command: enum4linux -a 192.168.56.110

```
(kali㉿kali)-[~]
└─$ enum4linux -a 192.168.56.110
Starting enum4linux v0.9.1 ( http://labs.portcullis.co.uk/application/enum4linux/ ) on Fri May 1 12:40:55 2026

===== ( Target Information ) =====
Target ..... 192.168.56.110
RID Range ..... 500-550,1000-1050
Username ..... ''
Password ..... ''
Known Usernames .. administrator, guest, krbtgt, domain admins, root, bin, none

===== ( Enumerating Workgroup/Domain on 192.168.56.110 ) =====

[+] Got domain/workgroup name: WORKGROUP

===== ( Nbtstat Information for 192.168.56.110 ) =====
```

```
Sharename      Type      Comment
-----
print$         Disk      Printer Drivers
tmp            Disk      oh noes!
opt            Disk
IPC$           IPC       IPC Service (metasploitable server (Samba 3.0.20-Debian))
ADMIN$         IPC       IPC Service (metasploitable server (Samba 3.0.20-Debian))
Reconnecting with SMB1 for workgroup listing.

Server         Comment
-----
Workgroup      Master
WORKGROUP      METASPLOITABLE

[+] Attempting to map shares on 192.168.56.110
//192.168.56.110/print$ Mapping: DENIED Listing: N/A Writing: N/A
//192.168.56.110/tmp Mapping: OK Listing: OK Writing: N/A
//192.168.56.110/opt Mapping: DENIED Listing: N/A Writing: N/A
```

Challenge 16 – Version Detection

Command: nmap -sV 192.168.56.110

```
(kali㉿kali)-[~]
$ nmap -sV 192.168.56.110
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-01 11:45 +08
Nmap scan report for 192.168.56.110
Host is up (0.0093s latency).
Not shown: 977 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rshd
513/tcp   open  login?
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs
: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://n
map.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 17.88 seconds
```

Challenge 17 – OS Detection

Command: `nmap -O 192.168.56.110`

```
(kali㉿kali)-[~]
$ nmap -O 192.168.56.110
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-01 11:57 +08
Nmap scan report for 192.168.56.110
Host is up (0.0025s latency).
Not shown: 977 filtered tcp ports (no-response)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
Warning: OSScan results may be unreliable because we could not find at least
1 open and 1 closed port
Device type: bridge|VoIP adapter|general purpose
Running (JUST GUESSING): Oracle Virtualbox (98%), Slirp (98%), AT&T embedded
(95%), QEMU (94%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:q
emu
Aggressive OS guesses: Oracle Virtualbox Slirp NAT bridge (98%), AT&T BGW210
voice gateway (95%), QEMU user mode network gateway (94%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.o
rg/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 9.72 seconds
```